

An implementation of the contextual bandit algorithm LinUCB

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Abstract. *This meta-paper presents the article selected for the project to be developed in the Artificial Intelligence course (CCM-102) at UFABC, first quarter of 2026.*

The reinforcement learning algorithm LinUCB, proposed by Li et al. [1] for personalized news recommendation, will be implemented and evaluated following the experimental protocol described in the original article.

An improved version of the algorithm will then be proposed, and both approaches will be compared.

Resumo. *Este meta-artigo apresenta o artigo selecionado para o projeto a ser desenvolvido na disciplina de Inteligência Artificial (CCM-102) da UFABC, no primeiro quadrimestre de 2026.*

O algoritmo de aprendizado por reforço LinUCB, proposto por Li et al. [1] para recomendação personalizada de notícias, será implementado e avaliado seguindo o protocolo experimental descrito no artigo original.

Uma versão aprimorada do algoritmo será então proposta, e ambas as abordagens serão comparadas.

1. Introduction

Bandit problems were introduced by Thompson W. R. [2] in order to determine whether one treatment of a certain class of individuals is better than a second based on the current data, without having to actually try it. The idea was to determine treatment allocation on the fly if a drug appears more or less effective.

The name Bandit comes from an animal learning experiment made in 1950s where a mice had to decide either to go left or right in a T-shaped maze. In one side it receives a reward, in other it receives nothing. To study it in humans, they made a similar setup with a "two-armed bandit" machine, where the person could pull left or right the arm of the machine, each giving a random pay-off with the distribution of pay-offs for each arm unknown to the player.

Major tech companies use bandit algorithms for configuring web interfaces, where applications include news recommendation, dynamic pricing and ad placement.

2. A Multi-armed Bandit Formulation

The article by Li et al. [1] formulates personalized news recommendation as a contextual bandit problem, a particular setting within reinforcement learning. In this framework, each interaction corresponds to a round or trial $t = 1, 2, 3, \dots$, in which the system observes contextual information and must select an action from a finite set A_t .

Li *et al.* define a three steps to be executed in each trial t :

1. Each action $a \in A_t$ is associated with a feature vector $x_{t,a}$ that represents the context at time t for both user u_t and arm a .
2. After selecting an action a_t , the algorithm observes only the reward r_{t,a_t} , while the rewards of non-selected actions remain unknown. This partial feedback characterizes the bandit setting.
3. The algorithm then improves its arm-selection strategy with the new observation $(x_{t,a_t}, a_t, r_{t,a_t})$. No feedback (payoff $r_{t,a}$) is observed for arms not chosen $a \neq a_t$.

With this, we have that the total T-trial payoff of the algorithm **A** is defined as $\sum_{t=1}^T r_{t,a_t}$. The idea of the algorithm is to maximize the total payoff. Other possibility is to minimize the total regret, defined by

$$R_A(T) \stackrel{\text{def}}{=} \mathbf{E} \left[\sum_{t=1}^T r_{t,a_t^*} \right] - \mathbf{E} \left[\sum_{t=1}^T r_{t,a_t} \right] \quad (1)$$

where a_t^* is the arm with maximum expected payoff at trial t .

In the context of article recommendation, we may consider the articles in the pool as arms. When an article is clicked by a user, a payoff of 1 is incurred; otherwise, the payoff is 0. Thus, the expected payoff of an article is its click-through rate (CTR). Choosing an article with maximum CTR is equivalent to maximizing the expected number of clicks from users, which in turn is the same as maximizing the total expected payoff in our bandit formulation.

3. Algorithm

The article will demonstrate an contextual bandit algorithm where the confidence interval can be computed efficiently in closed form when the payoff model is linear.

The confidence interval comes from a class of algorithms that is known as upper confidence bound algorithms (UCB). In a trial t , these algorithms estimate both the mean payoff $\hat{\mu}_{t,a}$ of each arm a as well as a corresponding confidence interval $c_{t,a}$, so that $|\hat{\mu}_{t,a} - \mu_a| < c_{t,a}$ holds with high probability.

The article calls the algorithm of **LinUCB** and is described by the pseudocode ??.

4. Dataset

Since the authors didn't provide the dataset used in the tests, we'll use the "News Click Behavior and Engagement Dataset" dataset, available in kaggle (<https://www.kaggle.com/datasets/zoya77/news-click-behavior-and-engagement-dataset>). This dataset contains information about articles and users, and will be used in the algorithms analyzed in this paper.

5. Improvement method

In order to improve the current method, we'll use another known method of recommendation system called Matrix factorization.

Algorithm 1 LinUCB with disjoint linear models

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1: Input:  $\alpha \in R_+$ 
2: for  $t = 1, 2, 3, \dots, T$  do
3:   Observe features of all arms  $a \in \mathcal{A}_t$ :  $\mathbf{x}_{t,a} \in R^d$ 
4:   for all  $a \in \mathcal{A}_t$  do
5:     if  $a$  is new then
6:        $A_a \leftarrow I_d$   $\triangleright d$ -dimensional identity matrix
7:        $b_a \leftarrow \mathbf{0}_{d \times 1}$   $\triangleright d$ -dimensional zero vector
8:     end if
9:      $\hat{\theta}_a \leftarrow A_a^{-1} b_a$ 
10:     $p_{t,a} \leftarrow \hat{\theta}_a^\top \mathbf{x}_{t,a} + \alpha \sqrt{\mathbf{x}_{t,a}^\top A_a^{-1} \mathbf{x}_{t,a}}$ 
11:  end for
12:  Choose arm  $a_t = \arg \max_{a \in \mathcal{A}_t} p_{t,a}$  (break ties arbitrarily) and observe payoff  $r_t$ 
13:   $A_{a_t} \leftarrow A_{a_t} + \mathbf{x}_{t,a_t} \mathbf{x}_{t,a_t}^\top$ 
14:   $b_{a_t} \leftarrow b_{a_t} + r_t \mathbf{x}_{t,a_t}$ 
15: end for
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Referências

- [1] L. Li, W. Chu, J. Langford, and R. E. Schapire. A Contextual-Bandit Approach to Personalized News Article Recommendation. In *Proceedings of the 19th International Conference on World Wide Web*, pages 661–670, Apr. 2010.
- [2] W. R. Thompson. On the Likelihood that One Unknown Probability Exceeds Another in View of the Evidence of Two Samples. *Biometrika*, 25(3/4):285, Dec. 1933.